



UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF ELECTRICAL ENGINEERING

SESSION 2025/2026 SEMESTER 2

COURSE CODE

MKEC 1123-01 – Advanced Microprocessor System

LECTURER'S NAME

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MILESTONE 1: STM32 FAMILIARIZATION

VIDEO LINK:

<https://youtu.be/5i81tzjD-fc>

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Setup Steps

Step 1: Create a New STM32 Project

Open STM32CubeIDE and navigate to File → New → STM32 Project. In the board selector, search for NUCLEO-F446RE, select it and click Next. Enter a project name and click Finish. When prompted, select Yes to initialize all peripherals with default configuration.

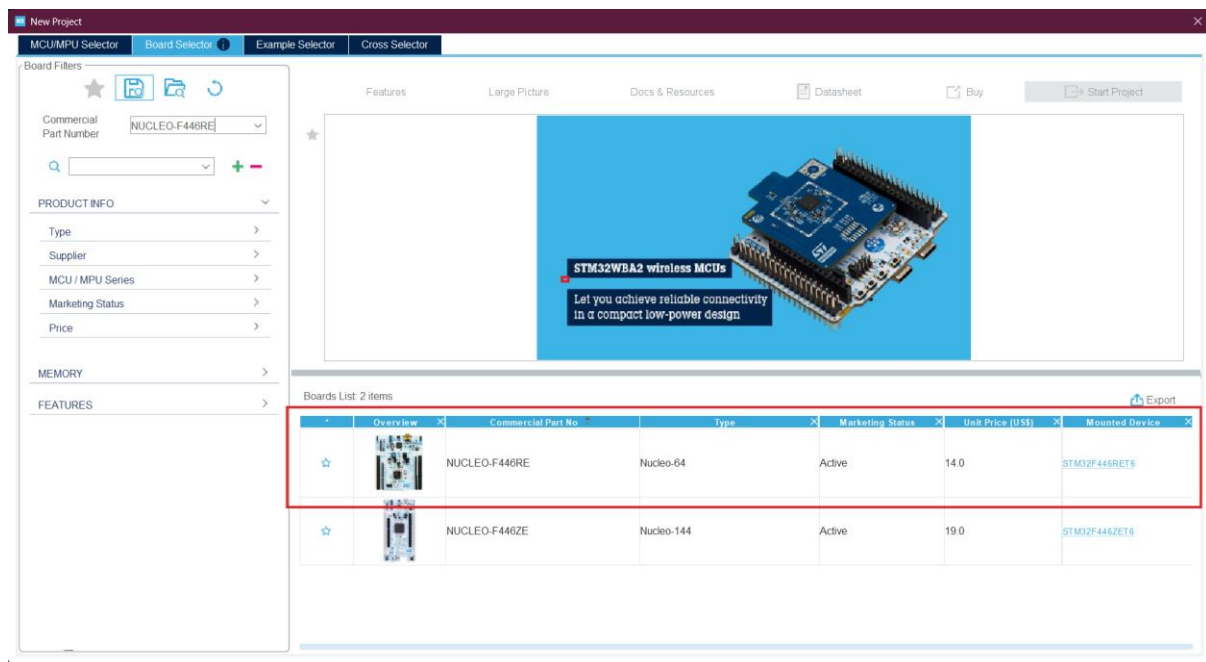


Figure 1 Board selection screen in STM32CubeIDE showing NUCLEO-F446RE selected

Step 2: Generate Code

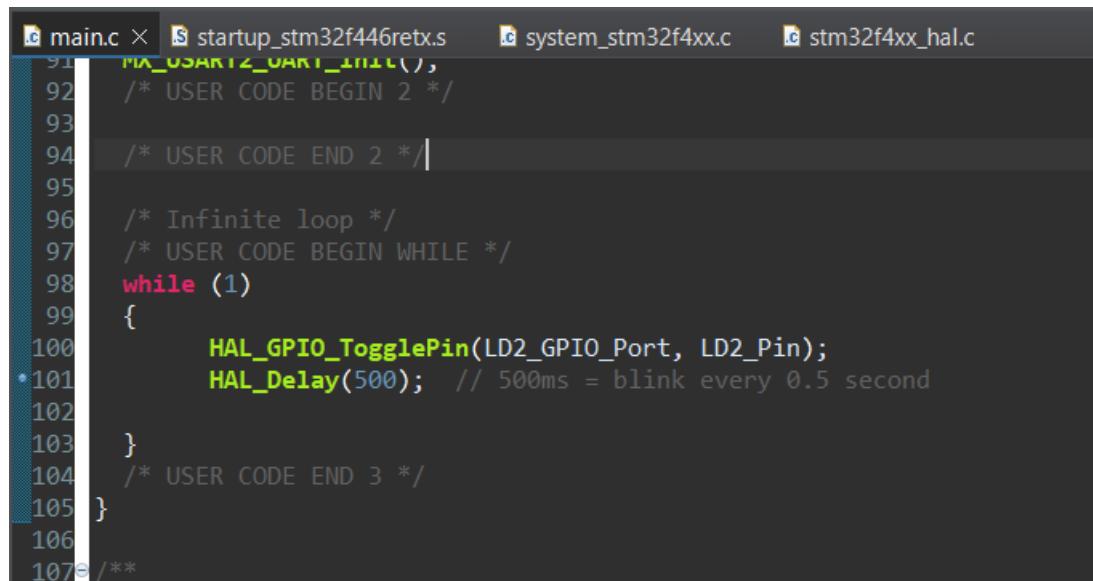
Click GENERATE CODE in the top right corner of CubeMX. When prompted, click Open Project to launch the generated project in STM32CubeIDE.

Step 3: Write the Blink Code

Open Core/Src/main.c. Inside the while(1) loop between the USER CODE BEGIN WHILE and USER CODE END WHILE comments, add the following code:

```
HAL_GPIO_TogglePin(LD2_GPIO_Port, LD2_Pin);
```

```
HAL_Delay(500); // 500ms = blink every 0.5 seconds
```



```
main.c × startup_stm32f446retx.s system_stm32f4xx.c stm32f4xx_hal.c
91 HAL_UART2_UART_Init();
92 /* USER CODE BEGIN 2 */
93
94 /* USER CODE END 2 */
95
96 /* Infinite loop */
97 /* USER CODE BEGIN WHILE */
98 while (1)
99 {
100     HAL_GPIO_TogglePin(LD2_GPIO_Port, LD2_Pin);
101     HAL_Delay(500); // 500ms = blink every 0.5 second
102 }
103 /* USER CODE END 3 */
104 }
105
106
107 /**
```

Figure 2 main.c showing the TogglePin and HAL_Delay code inside the while(1) loop

Step 4: Build, Flash and Run

1. Press Ctrl + S to save the file.
2. Press Ctrl + B to build. Confirm the console shows: Build Finished. 0 errors.
3. Press Ctrl + F11 → Run As → STM32 C/C++ Application to flash.
4. Wait for the console to show: Download verified successfully.
5. Press the black RESET button (B2) on the board.
6. Observe LD2 (green LED) blinking every 0.5 seconds.

```
Opening and parsing file: ST-LINK_GDB_server_a02492.srec

Memory Programming ...
  File       : ST-LINK_GDB_server_a02492.srec
  Size      : 8.42 KB
  Address   : 0x08000000

Erasing memory corresponding to segment 0:
Erasing internal memory sector 0
Download in Progress:

File download complete
Time elapsed during download operation: 00:00:00.392

Verifying...

File size < 32KB legacy verify will be used

Time elapsed during verifying operation: 00:00:00.055

Download verified successfully
```

Figure 3 Console output showing 'Download verified successfully' after flashing